Curriculum Vitae

Zhao-Wei Fu

E-mail Address fu.zhao.wei.fu.c8@tohoku.ac.jp

RESEARCH INTERESTS

- ✓ Corrosion and degradation behavior of nuclear structural materials in light water reactor environments.
- ✓ Microstructural optimization of austenitic stainless steel to enhance SCC resistance in high temperature water environment.
- ✓ Investigation of environmentally-assisted cracking mechanisms and mitigation strategies through advanced experimental techniques.
- ✓ Development of materials design approaches to extend the operational lifespan of nuclear power plants.

RESEARCH & WORK EXPERIENCE

Research Fellow, New Industry Creation Hatchery Center (NICHe), Tohoku University | Sendai, Japan October 2026 – Present

Research Assistant, Center for Energy & Environmental Research, NTHU | Hsinchu, Taiwan
June 2023 –August 2023

- Collect R&D literature and writing the research projects
- Set up the experimental facilities for the future research work in ECLab

Internship, Institute of Physics of Materials (IPM) | Brno, Czech Republic
October 2022 –December 2022

Project: Development of new type of oxide dispersion strengthened austenitic and ferritic alloys

- Fracture Brittle Group
- Advisor: Dr. Hynek Hadraba

Internship, Paul Scherrer Institut (PSI) | Villigen, Switzerland July 2022 – September 2022

Zinc project: *Effect of Zn injection on the SCC behavior of Alloy 182 and SS in LWR environments*

- Group of Component Structural Integrity (BTS)
- Advisor: Mr. Stefan Ritter, Mr. Hans-Peter Seifert
- Crack distribution analysis: evaluating and measuring cracks length distribution and density by Photoshop and Image J
- SCC cracking morphology analysis: using SEM to investigate the crack initiation sites
- Electrochemical measurement of oxide films: using EIS to obtain quantitative description of oxide films

Electrochemistry Lab (ECLab), NSE, NTHU | Hsinchu, Taiwan September 2020 – June 2022

Project:

[1]. Ministry of Science and Technology (MOST) Research Project

SCC initiation of Ni-based Alloys Under Supercritical Water Environment

[2]. National Tsing Hua University High Entropy Materials Center

Study of Electrochemical Properties of High-Entropy Alloys for Nuclear Structure Materials

- Characterization of oxide films of Ni-based superalloys and high-entropy alloys which were formed under the long-term exposure of supercritical water environment.
- Comparation of different oxidation behavior under different water chemistry
- Mechanism explanation of oxidation for different superalloys

EDUCATION

Department of Quantum Science and Energy Engineering, Graduate School of Engineering, Tohoku University | Sendai, Japan October 2023– September 2026

PhD student

Thesis title: *Improvement of Stress Corrosion Cracking Resistance of Austenitic Stainless Steels in High-Temperature Water via δ-Ferrite Incorporation*

- Proposing and demonstrating a new stainless steel microstructure with superior SCC resistance in high-temperature and high-pressure water

Institute of Nuclear Engineering and Science, National Tsing Hua University | Hsinchu, Taiwan September 2020 – January 2023

MSc

Thesis title: *Corrosion Behavior of High-entropy Alloys & Ni-based Superalloys in Supercritical Water Environments under Different Water Chemistry Conditions*

- Research Interests: High temperature water oxidation behavior of Ni-based alloys and HEAs

Department of Engineering and System Science, National Tsing Hua University | Hsinchu, Taiwan September 2016 – June 2020

BSc

- Research Interests: Electrochemical Analysis of stainless steels

PERSONAL SKILLS

Languages: Mandarin (Native), English (Proficient), Japanese (Intermediate)

Computer Skills: Image J, Python, Origin, Zview, Monte Carlo Simulation, SRIM/TRIM

Experiment & Equipment skills:

Instrument

- ✓ **SEM / EDS** – Proficient in microstructural and compositional analysis
- ✓ **Raman Spectroscopy** – Proficient in phase identification and surface characterization
- ✓ **EBSD** – Proficient in crystallographic orientation and grain boundary analysis
- ✓ **TEM** – Basic knowledge of microstructural observation
- ✓ **Electrochemical Analysis** – Proficient in polarization curve analysis and electrochemical impedance spectroscopy (EIS)
- ✓ **High-Frequency Induction Heating Furnace** – Experienced in thermal processing and microstructure modification

Experimental Techniques

- ✓ Setup and operation of a **recirculation loop system** for high-temperature and high-pressure water corrosion experiments
- ✓ Conducting **corrosion fatigue tests** under simulated LWR environments
- ✓ Application of **Direct Current Potential Drop (DCPD)** technique for in-situ crack length measurement
- ✓ **Fabrication of high-entropy alloys** through melting and solidification processes
- ✓ **Sample preparation** including grinding, polishing, and surface finishing
- ✓ **Post-corrosion treatment of specimens** including weighing, cutting, and mounting/embedding

SCIENTIFIC ACTIVITIES

Peer-reviewed Journal Publications

- **Zhao-Wei Fu**, Hiroshi Abe, Yutaka Watanabe, "Inducing δ-ferrite in 316NG austenitic stainless steel without welding: Microstructure and SCC resistance in simulated BWR environment," *Journal of Nuclear Materials*, vol. 621, p. 156383, Feb. 2026, doi: 10.1016/J.JNUCMAT.2025.156383.

Conference Proceedings

- **Zhao-Wei Fu**, Hiroshi Abe, Yutaka Watanabe, "Microstructural Design of Stainless Steels: Promoting δ-Ferrite Formation to Reduce SCC Propagation" *Proceedings of the 73rd Japan Conference on Materials and Environment ,* 2026, Paper No. A204. Japan Society of Corrosion Engineering.

Oral Presentations

- **Zhao-Wei Fu**, Hiroshi Abe, Yutaka Watanabe, "Development of stainless steel with excellent crack resistance due to island-shaped δ-ferrite at grain boundaries." *ICG-EAC 2026,* Idaho Falls, USA.

- **Zhao-Wei Fu**, Hiroshi Abe, Yutaka Watanabe, “Effect of Island-shaped δ-Ferrite along Grain Boundary on SCC Growth in 316NG Stainless Steels under simulated BWR environment.” *ICG-EAC 2025,* Le Pouliguen, France
- **Zhao-Wei Fu**, Mei-Ya Wang, Tsung-Kuang Yeh, “Oxidation Behavior of High-entropy Alloys & Ni-based Superalloys in Supercritical Water Environments under Different Water Chemistry Conditions.” *EUROCORR 2022,* Berlin, Germany.
- **Zhao-Wei Fu**, Mei-Ya Wang, Tsung-Kuang Yeh, “SCC Initiation for Nickel-based Alloy and High-entropy Alloy used in Supercritical Water Environment.” *EUROCORR 2021,* Virtual Conference.

GRANDS & AWARDS

Encouragement Award: Inducing δ-ferrite in 316NG austenitic stainless steel without welding: Microstructure and SCC resistance in simulated BWR environment Water Chemistry Division, Atomic Energy Society of Japan (AESJ)	*March 2026*
Peter Ford Grant ICG-EAC Annual Meeting	*May 2025*
Scholarship, Pioneering Research Support Project for PhD Students, Tohoku University, Japan	*October 2023-September 2026*
Young Member Excellence Award, American Nuclear Society, Taiwan Local Section	*August 2022*
Scholarship, Overseas Internship Program, Ministry of Education, Taiwan	*June 2022*
Scholarship, Berlin Paint Co., Ltd., The Corrosion Engineering Association of ROC	*November 2021*
The Excellent Award, 2021 Poster Competition of College of Nuclear Science, National Tsing Hua University	*September 2021*